Paper 4 Analysis Table

Category	Details
Research Paper Title	Advancing Breast Cancer-AI Diagnostics: An Explainable Deep Learning Model Using 2D Grayscale Ultrasound Imaging
Objective	To develop an explainable deep learning model for accurate classification of breast tumors (benign, malignant, normal) using ultrasound images.
Problem Statement	Breast cancer diagnosis using ultrasound is highly dependent on radiologists and may lead to inconsistent and subjective results, requiring automated and reliable AI solutions.
Model Used	- Deep Learning: MobileNetV2 (Transfer Learning) - Classification: 3-class (Benign, Malignant, Normal)
Programming / Tools	➢ Python ➢ TensorFlow / Keras ➢ Image preprocessing & augmentation ➢ t-SNE visualization
Dataset	BUSI Dataset (Breast Ultrasound Images) - 780 images - Benign, Malignant, Normal classes
Explainability	✔ Grad-CAM used to generate heatmaps highlighting tumor regions
Pros	✔ Lightweight model (MobileNetV2) ✔ High accuracy (~90%) ✔ Explainable AI integration (Grad-CAM) ✔ Suitable for low-resource environments
Cons	✘ Single dataset (BUSI only) ✘ Limited generalization ✘ Only grayscale images used ✘ No segmentation (only classification)
Challenges	⚠ Similarity between benign & malignant patterns ⚠ Limited dataset size ⚠ Model confusion in edge cases ⚠ Need for real-world validation
Conclusion	The proposed MobileNetV2-based model achieves strong classification performance while providing visual explanations, improving trust and usability in medical diagnosis.

Critical Comment	This paper improves over traditional models by adding explainability (Grad-CAM), but it is still limited to classification only. For your FYP (UltraXAI), you can enhance this by adding tumor segmentation (Swin-UNet) and advanced XAI techniques (Grad-CAM++, SHAP), making your system more powerful and clinically valuable.